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Artem Makoian

Programming: Python 3, C++
Languages: English (C1), Russian (native)
github.com/MakArtKar
scholar.google.com
leetcode.com/MakArtKar

Research Summary

Highlights
ICML 2026 paper on masked diffusion LMs: G-Star (poster)
Co-supervising 3 MSc theses as a 1st year PhD student
IOI national team candidate; 8th in the national Mathematics olympiad out of 1M participants

Topics	*Efficient LLM inference*	Discrete Diffusion, Flow Maps, Consistency, Inference-Time Scaling, Speculative Decoding
Skills	*Libraries*	PyTorch, Hugging Face Transformers (Qwen, LLaMA)
	Training / inference	Multi-node SLURM (DDP/FSDP), DeepSpeed, FlashAttention, bf16, vLLM/SGLang
	On-device inference	Pruning, quantization, ONNX
Background		5+ years in ML (engineer → researcher), PhD student; available to start anytime

Publications

Guided Star-Shaped Masked Diffusion — V. Meshchaninov, E. Shibaev, A. Makoian, et al. **ICML 2026**, poster
- Fine-tuning pretrained masked diffusion LMs for guided correction of already generated tokens

Motion ID: Human Authentication Approach — A. Gavron, et al., A. Makoian arXiv 2023
- User authentication by motion patterns: $82.2\% \pm 9\%$ accuracy (baseline 70%)

Research Experience

JetBrains **Feb 2025 — Present**
ML Researcher / JetBrains Research ***with D. Vetrov*** *Python 3, PyTorch, transformers, Hydra, Lightning, SLURM*

→Masked Diffusion LMs
- Inference-time scaling, diffusion drafters for speculative decoding, curriculum learning (Dream-7B, Qwen3-0.6B MDLM/BD3LM)

→Latent Diffusion Drafters for speculative decoding: lossless acceleration of LLM inference
- Distillation (flow maps, consistency) and latent space compression for real-time acceleration

AIRI (Artificial Intelligence Research Institute) **Jul 2022 — Oct 2022**
ML Research Intern / GAN adaptation *Python 3, PyTorch, Hydra, WandB*
- Few-Shot Image Generation implementation for few-shot GAN adaptation to a new domain from ∼10 images

Yandex **Jul 2021 — Oct 2021**
ML Researcher / VR Team ***with V. Lempitsky*** *Python 3, PyTorch*
- Trimap-free human alpha matting for VR scenes: improved FBA Matting, built alpha-mask datasets

Earlier: **Samsung R&D Department**, ML Research Intern, published Motion ID Nov 2020 — Jun 2021

Education

PhD in Computer Science (1st year), *Constructor University, Bremen, Germany* Sep 2026 — Present
- Teaching Assistant (seminars): Deep Bayesian Models (Spring 2026), Bayesian Methods (Fall 2026)
- Co-supervising 3 MSc theses, including Loop Transformers and Gaussian Diffusion LMs

Master in Computer Science, *Constructor University, Bremen, Germany* Sep 2024 — May 2026
Bachelor in Computer Science, *Neapolis University of Paphos, Cyprus*, GPA 86/100 Jan 2023 — Jun 2024
Bachelor in Computer Science, *HSE University, Moscow, Russia*, GPA 8.7/10 Sep 2019 — Jun 2024
Master's Equivalent in Data Science, *Yandex School of Data Analysis, Russia* Sep 2020 — May 2022

Industry Experience

Yandex **Jun 2023 — Sep 2024**
Middle ML Engineer / Geo services *Python 3, PyTorch, Hydra, OpenCV, LangChain*
- Road segmentation from aerial imagery with IoU 0.93; post-processing reduced artifacts from $10\% \to 0.02\%$
- News classification pipeline with LangChain agents for map update hypotheses: precision 0.9, recall 0.85

Sber Devices **Jul 2022 — Jun 2023**
Middle ML Engineer / Computer Vision Platform *Python 3, C++, PyTorch, ONNX, torch_pruning*
- Real-time background removal for video conferencing (0.98 Dice); pruning gave 1.5x speed-up
- On-device 700 KB payment card segmentation model with C++ inference in a multiplatform mobile SDK

Competitions

Candidate for the national team at the **IOI** (International Olympiad in Informatics) May, Nov 2018
Winner of the national olympiad in **Mathematics** (8th) and **Informatics** (33rd) out of 1M participants Apr 2018, 2019
Top 8% in Google Hash Code. **Top 50** in Google Kick Start. **Top 350 (top 0.8%)** in Google Code Jam Feb, May, June 2020